Automotive Safety: Complaints and Recall Intelligence

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Introduction

Vehicle safety recall is an important process in identifying and addressing safety defects in the automotive industry. This project investigates patterns in publicly available National Highway Traffic Safety Administration (NHTSA) Complaints and Recalls datasets. Vehicle complaints provide valuable information about safety issues reported by consumers, while recall records document actions taken to address identified defects. Analyzing these datasets together may reveal relationships between complaints and subsequent recalls, creating opportunities to improve vehicle safety.

Specifically, this project explores two primary questions through both supervised and unsupervised learning techniques. First, can information available at the time a complaint is filed help predict whether a recall will occur within the following two years? Second, can complaints be grouped into meaningful clusters that reveal patterns associated with higher rates of subsequent recalls? Through exploring these two questions, this project provides new perspectives on the potential value and limitations of NHTSA datasets, which can be further leveraged by vehicle manufacturers, regulators, and other automotive industry stakeholders to identify emerging safety concerns and prioritize investigations based on data-driven decision-making.

For supervised learning, we integrated complaint and recall records to evaluate several classification models. Among the representative models evaluated, K-nearest neighbors (KNN) achieved the strongest overall ranking performance. The tuned model identified approximately 53% of positive cases, although its precision remained relatively low. The results suggest that consumer complaints contain a useful but limited signal for anticipating subsequent recalls.

For unsupervised learning, we compared three clustering models trained on selected features including the sentence embeddings of complaint narratives. HDBSCAN identified the strongest associations with subsequent recalls, producing 45 clusters with recall rates of at least 70% on the full dataset, against a baseline rate of 43.7%. Its highest observed cluster recall rate was 96.2%. The results suggest that complaint narratives can be effective in identifying clusters among vehicle complaints.

The project is motivated in part by team members' experience in the automotive industry, which provides practical context for interpreting the data and understanding the operational significance of potential findings. At the same time, the project presents a substantial data-processing and machine-learning challenge. The datasets contain millions of records and are full of inconsistencies and unstructured information. Integrating these sources, processing their textual information, and establishing reliable relationships between complaints and recalls pose an exciting challenge.

Related Works

Several studies have analyzed publicly available datasets from NHTSA. We discuss three examples here for comparison.

First, *Analysis on the Trends and Characteristics of Vehicle Recalls in the United States* examines NHTSA recall data from 2010 to 2019, using descriptive statistics and Pearson correlation analysis to investigate recall trends and relationships between different recall characteristics. Second, *SafeRoad*, an application developed by a team at Virginia Tech, uses NHTSA complaint and recall data to predict whether complaints are likely to result in a vehicle recall. Using Naive Bayes classification and K-Nearest Neighbors (KNN), the application classifies complaints into two categories: recall or no recall. Third, *From Complaints to Clairvoyance: Deep Learning for Recall Prediction in the Automotive Industry* uses a BERT-based deep learning model to analyze complaint narratives and predict three possible investigation outcomes: no recall, a voluntary recall, or a mandatory recall.

Our proposal is a new project for our team, rather than a continuation of any prior coursework, and differs from these studies in several ways. First, whereas *Analysis on the Trends and Characteristics of Vehicle Recalls in the*

United States focuses on descriptive statistics and statistical relationships in historical recall data, our project uses supervised learning to examine the relationship between individual consumer complaints and subsequent recall outcomes. Second, while our project, like SafeRoad, also investigates the relationship between complaints and recalls, we explicitly enforce time-based training and testing to ensure that the model uses only information available before the prediction point. Third, although *From Complaints to Clairvoyance* shares our use of natural language processing (NLP) to analyze complaint narratives and supervised learning to predict recall-related outcomes, we further investigate complaint clusters through unsupervised learning, providing a complementary way to identify recurring narrative patterns associated with subsequent recalls.

Data Sources

Complaints

The Complaints dataset is publicly available through the NHTSA website. NHTSA also provides APIs for retrieving complaints by important variables such as vehicle make, model, model year, or Office of Defects Investigation (ODI) number. The downloadable dataset is provided in a raw text data format, while the API returns complaints information in JSON format. For this project, we use more than two million complaint records which serve as the primary input for both supervised and unsupervised learning.

Recalls

The Recall dataset is publicly available through the NHTSA website. NHTSA also provides APIs for retrieving recalls by important variables such as vehicle make, model, model year, or recall campaign number. The downloadable dataset is provided in a raw text data format, while the API returns recall information in JSON format. For this project, we use more than 245,000 recall records and integrate them with the NHTSA Complaints dataset.

VIN Decoder

For complaint records with unknown or missing model years, we use NHTSA's publicly available VIN Decoder to help identify vehicle information from the vehicle identification number (VIN). A VIN is a standardized 17-character identifier that encodes information about a specific vehicle including the model year.

Feature Engineering

The NHTSA Complaints and Recalls datasets were obtained as text files and converted into dataframes for analysis. We converted columns to appropriate data types, including integers for numerical fields and standardized datetime formats for date fields. Column names and text values were also normalized by removing leading and trailing whitespace and converting text to uppercase to improve consistency across the datasets.

During the initial data exploration, we observed that the original Recalls dataset contained records related to vehicles, equipment, and other product types. Because this project focuses on motor vehicles, we filtered the dataset to retain vehicle-related recalls. We also examined the temporal coverage of the Complaints dataset and restricted the data as needed. This decision was motivated by changes in automotive safety regulations and vehicle technologies over time, which could make older complaint patterns less representative of more recent safety issues. For records with missing information, such as model year, we investigated whether values could be recovered using vehicle identification numbers (VINs).

We then standardized component descriptions across the two datasets. The COMPDESC field in the Complaints dataset contained more than 700 unique values, often organized hierarchically into general and specific component descriptions. The COMPNAME field in the Recalls dataset presented a similar challenge. To reduce the number of categories while retaining meaningful distinctions, we reviewed NHTSA component classification resources and other automotive references, including the Universal Technical Institute's categorization of automotive systems and parts. Based on these references, we developed a project-specific taxonomy consisting of 16 major component categories.

After cleaning and standardizing the datasets, we engineered additional features where appropriate, including complaint counts and reported injuries. We also integrated the Complaints and Recalls datasets to investigate relationships between consumer-reported problems and subsequent recalls. The merged data showed that many recall campaigns had no associated consumer complaints in the available dataset. This suggests that some defects may be identified through manufacturer investigations or other channels rather than consumer complaints alone.

These shared preprocessing steps established a consistent foundation for the supervised and unsupervised learning pipelines, which were then developed separately to address their respective objectives.

Supervised Learning Feature Processing

For supervised learning, the preprocessing pipeline was designed to transform the Complaints and Recalls datasets into a merged dataset suitable for classification. To generate classification labels, we constructed matching keys based on vehicle make, model, model year, and component category. Using a nearest-match join, we identified the first matching recall occurring strictly after each complaint and the most recent matching recall occurring on or before it.

To improve label reliability, we excluded complaints filed before January 2010, for which earlier recall history might be unavailable, and complaints filed after September 2024, for which the two-year follow-up period was incomplete. Although these records were excluded from model training and evaluation, they remained available for calculating historical features.

We then divided the remaining data chronologically, assigning the most recent 20% of complaints to the test set and the most recent 15% of the remaining records to the validation set. A 730-day gap was introduced before each evaluation period to reduce overlap between the two-year label windows of earlier records and subsequent data splits. This approach was intended to better represent real-world deployment, in which models are trained on historical complaints and used to predict future recall outcomes.

Numerical features were imputed using medians calculated from the training set, with additional binary indicators introduced to identify missing values. The resulting features were standardized to have a mean of zero and a standard deviation of one. Categorical features were one-hot encoded, with categories occurring fewer than 30 times in the training data grouped into an infrequent category. Categories appearing for the first time in the validation or test sets were handled consistently to prevent mismatches across datasets.

Complaint narratives were transformed into numerical representations using TF-IDF, incorporating individual words and two-word phrases while limiting the vocabulary to 20,000 terms that appeared in at least five complaints and removing English stop words. The resulting sparse text representation was used directly by linear models. Random forest and KNN instead received a reduced representation generated through truncated singular value decomposition (SVD), which compressed the text features into 100 dimensions to make the data more manageable for these models.

[Table 1: Supervised Learning Features]

Feature	Description
CRASH	1 if the complaint's CRASH field is Y, indicating that the incident involved a crash; otherwise, 0.
FIRE	1 if the complaint's FIRE field is Y, indicating that the incident involved a fire; otherwise, 0.
POLICE_RPT_YN	1 if a police report was filed (POLICE_RPT_YN = Y); otherwise, 0.
ORIG_OWNER_YN	Indicates whether the original owner reported
ANTI_BRAKES_YN	1 if the vehicle has anti-lock brakes (ANTI_BRAKES_YN = Y); otherwise, 0.
CRUISE_CONT_YN	1 if the vehicle has cruise control (CRUISE_CONT_YN = Y); otherwise, 0.
MEDICAL_ATTN	1 if medical attention was sought (MEDICAL_ATTN = Y); otherwise, 0.
VEHICLES_TOWED_YN	1 if the vehicle had to be towed (VEHICLES_TOWED_YN = Y); otherwise, 0.
REPAIRED_YN	Indicates whether the tire was repaired
MAKETXT	Vehicle make
MODELTXT	Vehicle model
COMPONENT	Vehicle component associated with the report
CMPL_TYPE	Source of the consumer complaint
STATE	U.S. state associated with the complaint
DRIVE_TRAIN	Vehicle drivetrain configuration
FUEL_TYPE	Fuel source used by the vehicle
TRANS_TYPE	Vehicle transmission type
vehicle_age	Age of the vehicle
INJURED	1 if the INJURED count is greater than 0; otherwise, 0.

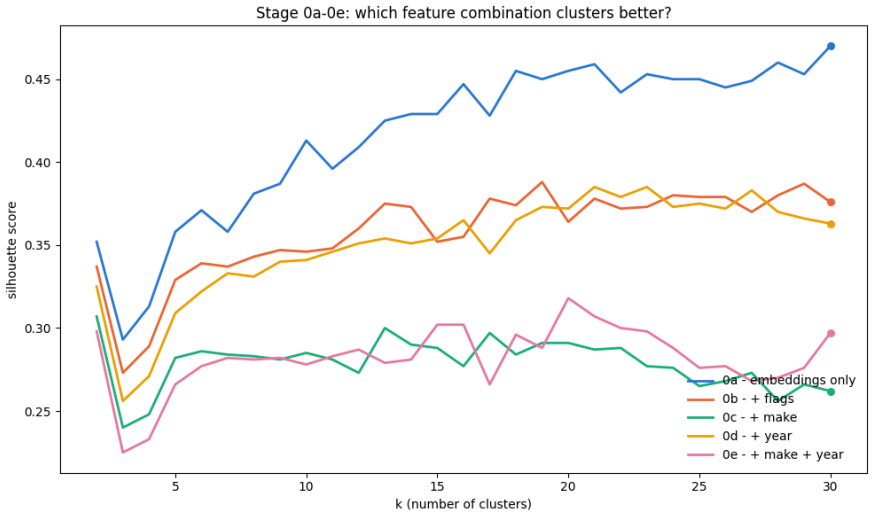
DEATHS	1 if the DEATHS count is greater than 0; otherwise, 0.
MILES	Vehicle mileage
OCCURENCES	Number of reported occurrences
VEH_SPEED	Vehicle speed
text_len	Length of the complaint text
days_fail_to_report	Number of days between the incident and its report
prior_recall_same_group	Indicates whether a matching recall occurred
days_since_prior_recall	Number of days elapsed since the relevant previous recall
grp_prior_total	Total number of relevant prior complaints
grp_prior_90d	Number of relevant prior complaints during the preceding 90 days.
grp_prior_365d	Number of relevant prior complaints during the preceding 365 days.
grp_crash_fire_365d	Number of relevant prior complaints during the preceding 365 days with a crash or fire
mmy_prior_365d	Number of relevant prior complaints during the preceding 365 days for the same make, model, and year
make_comp_prior_365d	Number of relevant prior complaints during the preceding 365 days for the same make and component

Unsupervised Learning Feature Processing

For unsupervised learning, several feature representations were explored to determine which features best supported clustering quality. Sentence embeddings generated using the all-MiniLM-L6-v2 model were selected to capture the semantic content of vehicle complaints, with each complaint represented by a 384-dimensional vector reduced to 10 dimensions using UMAP.

On the 30,000-record sample, K-means achieved its highest silhouette score at k=20 when using sentence embeddings alone, with a score of 0.455. Adding structured features reduced clustering quality, with silhouette scores of 0.364 for embeddings plus severity and equipment flags, 0.372 for embeddings plus model year, 0.318 for embeddings plus make and year, and 0.291 for embeddings plus make. Sentence embeddings also consistently outperformed the other feature combinations across the tested range of k=2 to k=30.

[Chart 1: Stage 0]



These results suggested that the semantic content of complaint narratives provides a more effective basis for clustering than the additional vehicle attributes evaluated. Therefore, sentence embeddings alone were retained as the final feature representation for subsequent clustering experiments.

[Table 2: Unsupervised Learning Features]

Feature	Description
emb_0–emb_383	The 10-dimensional sentence embedding of the complaint's narrative (CDESCR).
crash_flag	1 if the complaint's CRASH field is Y, indicating that the incident involved a crash; otherwise, 0.
fire_flag	1 if the complaint's FIRE field is Y, indicating that the incident involved a fire; otherwise, 0.
injured_flag	1 if the INJURED count is greater than 0; otherwise, 0.

death_flag	1 if the DEATHS count is greater than 0; otherwise, 0.
police_report_flag	1 if a police report was filed (POLICE_RPT_YN = Y); otherwise, 0.
medical_attn_flag	1 if medical attention was sought (MEDICAL_ATTN = Y); otherwise, 0.
towed_flag	1 if the vehicle had to be towed (VEHICLES_TOWED_YN = Y); otherwise, 0.
anti_brakes_flag	1 if the vehicle has anti-lock brakes (ANTI_BRAKES_YN = Y); otherwise, 0.
cruise_control_flag	1 if the vehicle has cruise control (CRUISE_CONT_YN = Y); otherwise, 0.
mk	The vehicle's make. This feature is excluded from the clustering input.
md	The vehicle's model. This feature is excluded from the clustering input.
yr	The vehicle's model year. This feature is excluded from the clustering input.
DATEA	The date the complaint was filed.
PARENT_GROUP	The complaint's component category.. This feature is excluded from the clustering input.

Part A. Supervised Learning

Methods Description

Several classification models were trained and evaluated to compare their performance on the recall prediction task. However, this section focuses on four representative models that provide different approaches to classification. The dummy classifier serves as a baseline against which the performance of more sophisticated models can be evaluated. Logistic regression was selected as a linear classification model because of its computational efficiency and suitability for high-dimensional, sparse text data. Random forest was included to capture nonlinear relationships among features by combining multiple decision trees trained on different bootstrap samples of the training data. K-nearest neighbors (KNN) was also evaluated as a distance-based classification method that predicts the class of an observation based on the labels of its nearest training examples. Together, these models provide a basis for comparing baseline performance, linear classification, nonlinear ensemble learning, and distance-based classification.

Supervised Evaluation

Overall Results

The target variable was imbalanced, with positive cases representing a relatively small proportion of the observations. In the test period, 10.2% of complaints were followed by a matching recall within two years, compared with 8.6% in the training period. This imbalance influenced the selection of evaluation metrics, as accuracy alone would not provide a meaningful assessment of model performance. For example, a model that predicts no recall for every complaint would achieve approximately 90% accuracy in the test set despite failing to identify any positive cases. Therefore, we used several complementary metrics to evaluate the models.

Precision-recall area under the curve (PR-AUC), reported as average precision, was selected as the primary metric because it evaluates the trade-off between precision and recall across different classification thresholds. ROC-AUC was also used to measure the model's ability to rank positive cases above negative cases, with a value of 0.5 representing random ranking. In addition, precision, recall, and F1-score were calculated at a decision threshold selected to maximize F1 on the validation set and then applied unchanged to the test set.

To assess the models' ability to prioritize complaints for further investigation, we also measured precision among the highest-ranked 1% and 5% of complaints, representing the proportion of actual positive cases within these groups.

Finally, the Brier score was used to evaluate the accuracy of the predicted probabilities, with lower values indicating better probabilistic predictions. To maintain a consistent evaluation procedure and avoid information leakage, each model was fitted using the training set, while the validation set was used for hyperparameter selection and decision-threshold selection. The final performance metrics were calculated on the test set, which was reserved for the final evaluation.

Overall, KNN took the lead on most ranking metrics.

[Table 3: Supervised Learning Model Performances]

Model	PR-AUC	ROC-AUC	F1	Precision	Recall	Precision@1%	Precision@5%	Brier
KNN	0.233	0.755	0.341	0.249	0.538	0.118	0.299	0.09
Random forest	0.204	0.732	0.198	0.270	0.155	0.090	0.263	0.09
Logistic regression	0.149	0.630	0.190	0.178	0.205	0.180	0.169	0.15
Dummy (baseline)	0.102	0.500	0.185	0.102	1.000	0.103	0.105	0.09

In-Depth Evaluation

Hyperparameter tuning was conducted to evaluate whether model performance could be improved by selecting more suitable parameter settings. For each model, 20 random configurations were sampled from a predefined search space, trained on the training set, and evaluated using PR-AUC on the validation set.

[Table 4: Supervised Learning Model Settings]

Model	Settings searched
Random forest	trees (300–800), maximum depth (15 to unlimited), minimum rows per leaf (1–50), features per split (square root, 5–20%), class weighting
KNN	neighbours (10–400), uniform or distance weighting, Euclidean or cosine distance

However, the results varied between models. Random forest showed signs of overfitting to the validation set, with PR-AUC decreasing from 0.241 during validation to 0.181 on the test set, which was also lower than its performance using default settings.

In contrast, K-nearest neighbors (KNN) showed little improvement from hyperparameter tuning. Its test PR-AUC decreased slightly from 0.233 with default settings to 0.226 after tuning, while validation and test performance remained relatively similar at 0.220 and 0.226, respectively. This consistency suggests that KNN generalized more reliably across the two data splits, although tuning did not produce a meaningful improvement in its overall performance.

Ablation analysis (KNN)

To examine the contribution of different feature groups to model performance, an ablation study was designed using KNN. This approach involves removing one feature group at a time and measuring the resulting change. A larger decrease in performance would suggest that the model relies more heavily on the removed feature group.

[Table 5: Supervised Learning KNN Ablation Results]

Run	Features removed	What it tests	Test PR-AUC	Change vs full	Precision@5%
Full model	none	Tuned KNN model at its best setting	0.233	—	0.299
no_make_model	MAKETXT, MODELTXT	Does the model just memorize risky makes and models?	0.219	−0.0004 (−0.2%)	0.259
no_vehicle_identity	MAKETXT, MODELTXT, mmy_prior_365d, make_comp_prior_365d	Make/model plus the complaint counts grouped by make/model	0.221	+0.0022 (+1.0%)	0.253
no_component	COMPONENT	The component category	0.223	+0.0039 (+1.8%)	0.271
no_history_counts	grp_prior_total, grp_prior_90d, grp_prior_365d, grp_crash_fire_365d, mmy_prior_365d, make_comp_prior_365d	How many similar complaints came before this one	0.214	−0.0048 (−2.2%)	0.330
no_prior_recall	Prior_recall_same_group, days_since_prior_recall	Whether this vehicle and component had already been recalled	0.174	−0.0446 (−20.3%)	0.211
no_cmpl_type	CMPL_TYPE	Leakage check on the complaint-type code	0.214	−0.0054 (−2.5%)	0.248
no_severity	CRASH, FIRE, INJURED, DEATHS, MEDICAL_ATTN, POLICE_RPT_YN, VEHICLES_TOWED_YN	Crash, fire, injury and death signals	0.221	+0.0024 (+1.1%)	0.25
no_text	CDESCR (text)	How much the narrative adds	0.218	−0.0009 (−0.4%)	0.252
text_only	all structured features	How well the narrative does on its own	0.101	−0.1180 (−53.9%)	0.102

The ablation analysis indicates that prior recall information contributed most to KNN's predictive performance, with PR-AUC decreasing from 0.233 to 0.174 when the feature was removed. In contrast, excluding make/model information, component categories, severity indicators, or complaint narratives led to relatively small changes. The text-only model achieved a PR-AUC of 0.101, substantially lower than the full model, suggesting that complaint narratives alone provide limited predictive value.

Sensitivity analysis

The sensitivity analysis was designed to assess whether KNN's performance advantage remained consistent under reasonable changes to the modelling approach. Four design choices were varied individually to examine their potential influence on predictive performance.

[Table 6: Supervised Learning KNN Sensitivity Results]

Choice varied	Values	Command change	Test PR-AUC	Lift over that run's dummy
Look-ahead horizon	180/ 365 / 730 (main) days	--horizon-days 365 (or 180)	180: 0.086 365: 0.147 730: 0.219	180: 2.02x 365: 2.14x 730: 2.18x
Neighbours (k)	10 / 25 / 50 / 100 / 200 / 400	read from outputs/tuning/trials_knn.csv	Validation PR-AUC 10 : 0.176, 25 : 0.205 50 : 0.213, 100 : 0.215 200 : 0.216, 400 : 0.220	N/A

The sensitivity analysis shows that KNN's predictive performance improved as the look-ahead horizon increased, with test PR-AUC rising from 0.086 at 180 days to 0.219 at 730 days. Increasing the number of neighbours from 10 to 400 resulted in gradual improvements in validation PR-AUC, from 0.176 to 0.220, although the gains became smaller at higher values. Overall, the results suggest that the prediction horizon has a substantial influence on performance, while increasing the number of neighbours provides more modest improvements.

Identify Tradeoffs

Overall, while KNN provided the strongest general performance, no single model performed best across every metric, highlighting the importance of selecting a model based on the intended application.

One of the most important tradeoffs is between precision and recall, particularly because only 10.2% of complaints in the test set were followed by a matching recall within two years. KNN identified 53.8% of actual positive cases, the highest among the evaluated models. However, its precision of 24.9% suggests a high level of false positives. Random forest's precision of 27.0% is the highest among the trained models, but its recall of 15.5% means it misses most positive cases. KNN would therefore be more suitable when the goal is to identify as many potentially relevant complaints as possible, whereas random forest may be slightly preferable when the cost of investigating false positives is particularly high.

Further, the hyperparameter tuning results highlight a tradeoff between improving validation performance and maintaining performance on unseen data. KNN's default test PR-AUC was 0.233, compared with 0.226 after tuning. The tuned model therefore did not improve test performance, although its validation and test results were relatively close. This suggests that more extensive tuning does not necessarily lead to a better-performing model.

Failure analysis

At the selected decision threshold of 0.148, tuned KNN achieved a precision of 26.6% and a recall of 52.8% on the test set. To illustrate these results, of every 1,000 test complaints, approximately 203 would be flagged as likely to be followed by a recall within two years, of which 54 would be true positives and 149 would be false positives. These results indicate that false positives outnumber true positives by approximately three to one, highlighting the limitations of using consumer complaints alone to predict subsequent recalls.

Several factors may explain why some complaints followed by recalls are missed by the model.

First, the model relies heavily on historical complaint counts and prior recall signals, making it difficult to identify defects that emerge without a preceding pattern of consumer complaints. For example, a complaint about a 2020 GMC Acadia's intermittent rearview camera failure received a low KNN score of 0.023, below the 0.190 risk threshold, despite a recall being issued 51 days later following defect investigations. The complaint had no prior reports for the same vehicle and component, no earlier recall, and no reported crash or injury. Similar issues may be avoided by adding data sources beyond complaints, such as NHTSA's investigation records, technical service bulletins and warranty extensions to capture recalls driven by regulators or manufacturers rather than consumer complaints.

Second, the fixed 730-day prediction horizon can cause complaints associated with eventual recalls to be labeled as false alarms when the recall occurs just outside the window. For example, a complaint about unintended braking in a 2018 Tesla Model S received a high KNN risk score of 0.293, exceeding the 0.190 threshold, but was labeled as no recall because the matching recall was issued 733 days later. The model identified meaningful warning signals, including 19 earlier complaints for the same vehicle and component and an earlier recall, yet this complaint was labeled incorrectly. In future iterations, we could consider scoring near-misses with a graded or time-weighted label so that a recall at 733 days would count similar as the one at 729 days to address similar issues.

Third, high complaint volume does not necessarily indicate that a vehicle defect will lead to a safety recall, potentially causing false alarms for heavily complained-about vehicles. For example, a complaint about severe steering-wheel shaking in a 2019 Ford F-250 Super Duty received a KNN risk score of 0.203, exceeding the 0.190 threshold, despite no steering-related recall being issued. The model relied on substantial historical complaint activity, including 650 earlier complaints for the same vehicle and component. However, complaint volume alone does not predict whether a defect will lead to a safety recall. Such issues can be avoided by considering additional information such as NHTSA's investigation outcomes and giving more weight to the details of the severity, loss of control, and safety hazards.

Overall, these findings suggest that consumer complaints contain a meaningful but limited signal for predicting subsequent recalls. KNN identifies slightly more than half of the positive cases, but its relatively low precision means that many flagged complaints are not followed by a matching recall within the specified period. The model may therefore be more useful for prioritizing complaints for further investigation than for making reliable recall predictions independently. Further analysis of the missed cases and false positives would help determine whether incorporating additional information, such as vehicle exposure, manufacturer investigations, or warranty data, could improve predictive performance.

Part B. Unsupervised Learning

Methods description

Three clustering algorithms were selected and evaluated to explore potential patterns in vehicle complaints. First, K-means was included as a centroid-based clustering method that assigns each data point to one of a predefined number of clusters by minimizing within-cluster variance. Second, HDBSCAN was selected as a density-based, hierarchical method that identifies clusters based on regions of high point density and labels points in low-density regions as noise rather than forcing them into clusters. Finally, agglomerative clustering was also included as a hierarchical method that builds a dendrogram by iteratively merging the closest clusters, with the final grouping determined by a chosen number of clusters. Together, these algorithms provide a basis for comparing centroid-based, density-based, and hierarchical approaches to identifying patterns in vehicle complaints.

The experiment was then organized into four stages:

Stage 0 — Feature Selection evaluates the input representation, including the UMAP dimensionality and whether the structured features improve clustering quality. K-means is used as the primary model for these experiments.

Stage 1 — Parameter tuning identifies suitable hyperparameters for each clustering algorithm.

Stage 2 — Full-Dataset Training trains the final models using the feature representation and hyperparameters selected in the earlier stages. K-means and HDBSCAN are trained on the full dataset of approximately 2.25 million complaints, while agglomerative clustering is applied to a sample of 50,000 complaints.

Stage 3 — Validation evaluates the resulting clusters against subsequent NHTSA recalls to determine whether the identified complaint patterns are associated with recalls that occurred after the complaints were reported.

Several evaluation metrics were used to assess the clustering results and compare the three algorithms. The silhouette score was used for K-means and agglomerative clustering as a standard internal validation metric that measures how well each data point fits within its assigned cluster compared to the nearest alternative cluster. For HDBSCAN, the Density-Based Clustering Validation (DBCV) score was used because it accounts for differences in cluster density and accommodates noise points, which are explicitly identified by HDBSCAN. In addition to these internal metrics, a project-specific recall rate per cluster was developed to evaluate the relevance of the identified clusters to vehicle safety recalls. For each cluster, this metric measures the proportion of complaints associated with a vehicle make, model, and year that subsequently had an NHTSA recall dated after the complaint.

[Table 7: Unsupervised Learning Models & Parameters Tuning]

Model	Parameter	Candidates	Chosen	Evidence
(shared) UMAP	n_components	5, 10, 15, 20, 30	10	Silhouette 0.436 / 0.455 / 0.453 / 0.455 / 0.440; 10 tied for best and was cheaper than 20
K-means	k	15, 18, 20, 22, 25 × 5 seeds	20	Mean silhouette 0.456, std 0.004 — near-top mean, lowest spread
HDBSCAN	min_cluster_size (min_samples=10)	200, 500, 1000, 2000	500	DBCV 0.171 → 0.149 across the range; 500 halves the cluster count vs 200 (122 → 62) for 0.008 DBCV
HDBSCAN	min_samples (min_cluster_size=500)	5, 10, 15, 25	25	DBCV 0.151 / 0.163 / 0.166 / 0.176; noise 35.4% → 31.9%; cluster count flat (56–64)
Agglomerative	linkage (n_clusters=20)	ward, average, complete, single	ward	Most balanced (largest/smallest cluster 127x vs ~540x); single linkage degenerate (silhouette -0.227)
Agglomerative	n_clusters (linkage=ward)	5–40	20	Silhouette plateau 0.433–0.441 for n=16–21; past n=22 gains come from splitting off 12–15 point clusters

Unsupervised Evaluation

Overall Results

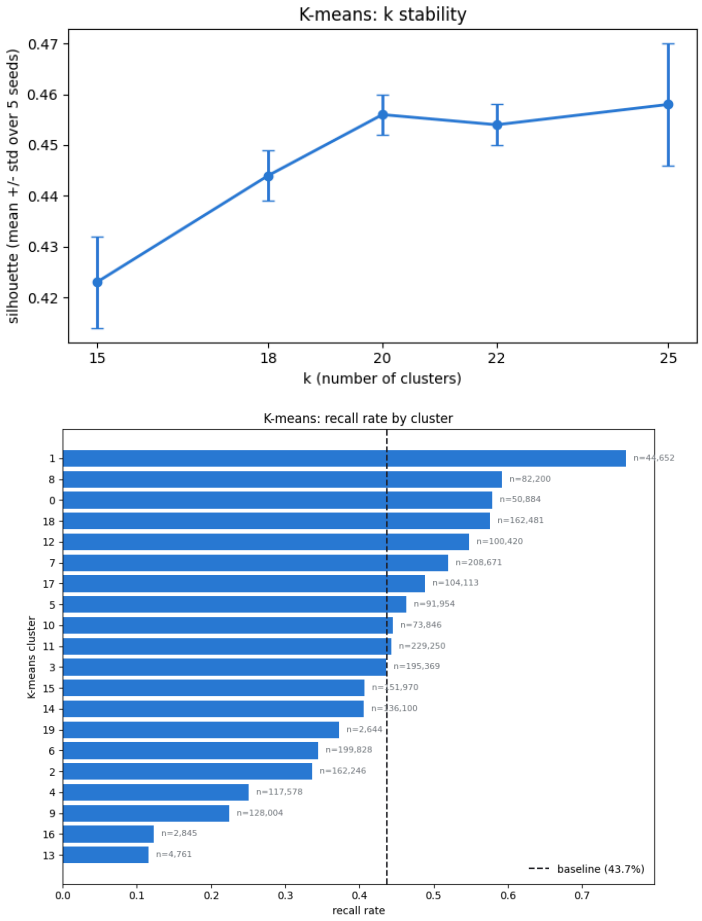
HDBSCAN identified the largest number of clusters with recall rates above baseline, including dozens of clusters with strong associations with subsequent recalls. Its highest recall rate was 96.2%, compared with 76.0% for K-means and 76.2% for agglomerative clustering. These peak values, however, should be interpreted with caution because HDBSCAN produced 278 clusters, while K-means produced 20 larger clusters.

The higher peak for HDBSCAN may partly reflect the smaller size of some clusters, as the sensitivity analysis showed that the highest recall rate increased as the minimum cluster size decreased. Therefore, the number of clusters with recall rates of at least 70% provides a more useful basis for comparison, with HDBSCAN identifying 45 such clusters. Given the number of clusters produced by HDBSCAN, the report focuses on the top 10–20 clusters by recall rate, with each selected cluster containing at least 500 complaints.

[Table 8: Unsupervised Learning Model Performances]

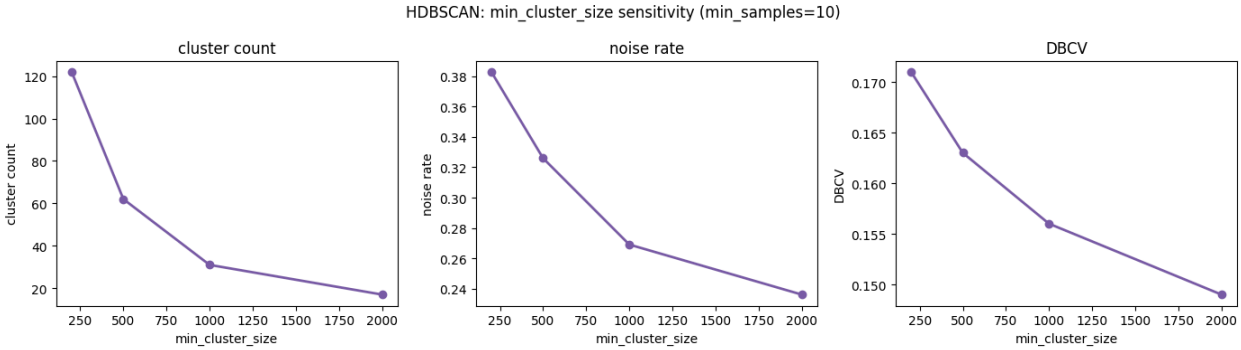
Model	Trained On	Baseline	Strongest Credible Finding	Notes
K-means (k=20)	Full 2,249,816 rows	43.7%	cluster 1: 76.0% recall rate (1.74x baseline), n=44,652	Visualization 1
HDBSCAN (min_cluster_size=500, min_samples=25)	Full 2,249,816 rows	43.7%	cluster 245: 96.2% (2.20x baseline), n=1,460 cluster 195: 95.1%, n=4,094	Visualization 2
Agglomerative (linkage=ward, n_clusters=20)	50,000-row sample	44.1%	cluster 16: 76.2% (1.73x baseline), n=949	Visualization 3

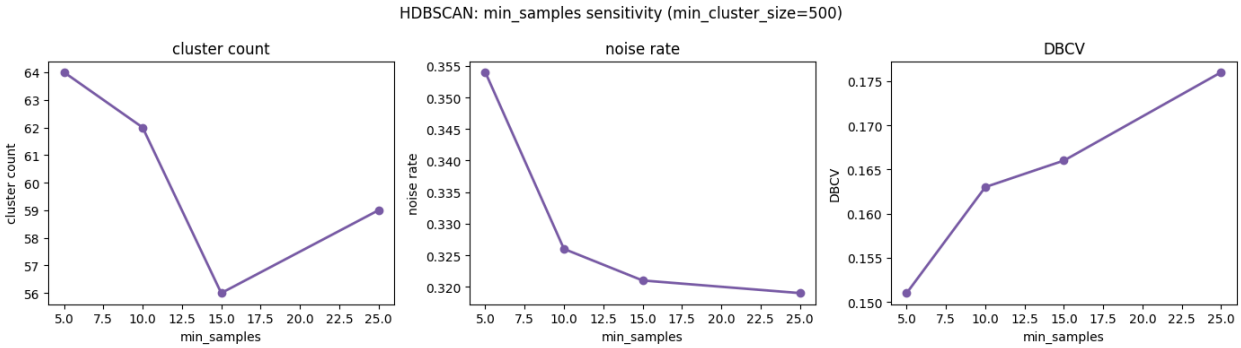
[Chart 2: K-means]

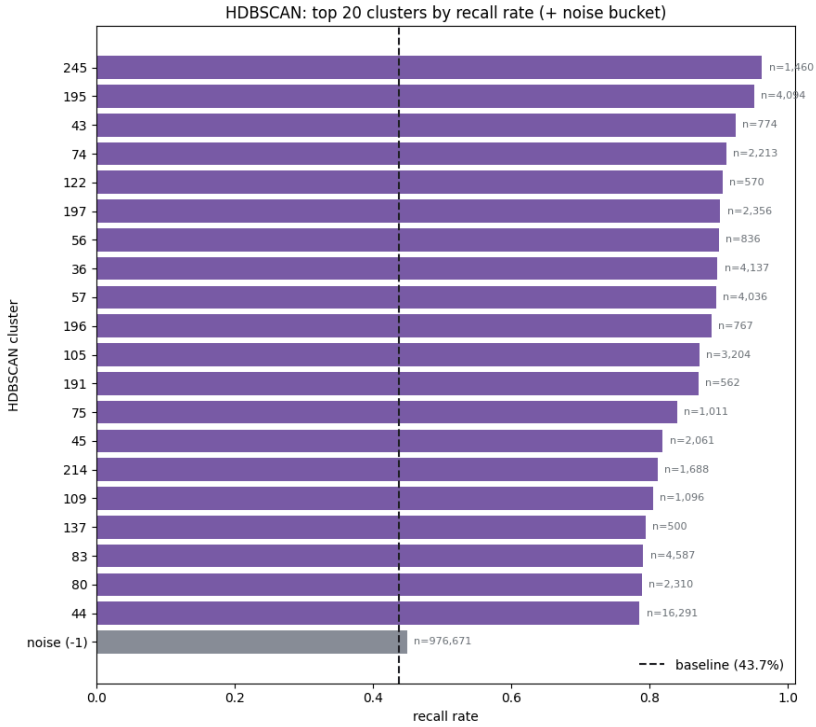


K-means stability was evaluated by calculating the mean silhouette score and standard deviation for each value of k across five random seeds. The results showed that a small number of clusters had recall rates between 52% and 59%, while the remaining clusters ranged from 11% to 49%. The lowest recall rates were 11.6% for a cluster containing 4,761 complaints and 12.3% for a cluster containing 2,845 complaints. These findings suggest that the strength of the association with subsequent recalls varied considerably across clusters.

[Chart 3: HDBSCAN]

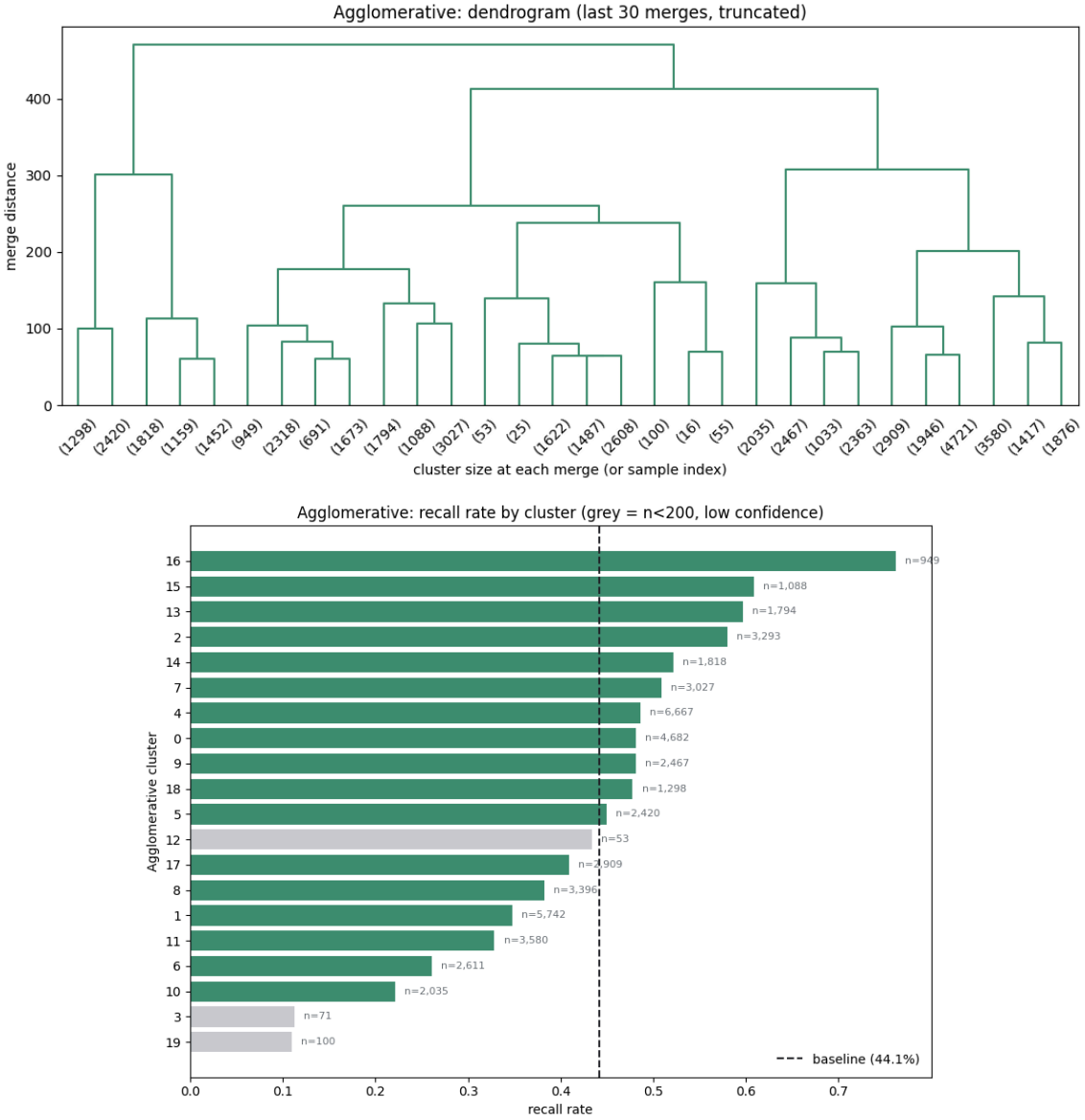






HDBSCAN identified 45 clusters with recall rates of at least 70%, including 16 clusters above 80% and six above 90%. All clusters contained at least 500 complaints, providing a substantial number of observations for evaluating their association with subsequent recalls. However, recall rates varied considerably across clusters, with the lowest rates being 0.1% for a cluster containing 853 complaints, 0.4% for a cluster containing 815 complaints, and 2.1% for a cluster containing 1,310 complaints. HDBSCAN also classified 976,671 complaints, representing 43.4% of the dataset, as noise. This group had a recall rate of 44.9%, which was approximately equal to the overall baseline. This suggests that while HDBSCAN identified several clusters strongly associated with subsequent recalls, a substantial portion of complaints did not belong to a distinct cluster with a stronger recall association.

[Chart 4: Agglomerative]



Agglomerative clustering identified eleven clusters with recall rates above the baseline, including one exceeding 70%. Most clusters contained several thousand complaints, providing a substantial number of observations for evaluating their association with subsequent recalls. However, two small clusters, containing 71 and 100 complaints, respectively, had recall rates of approximately 11% and 10%, and were considered low-confidence results. Overall, the results suggest that agglomerative clustering can identify groups of complaints with recall rates substantially above the baseline, although recall rates differ considerably across clusters and the quality of the resulting segmentation depends strongly on the linkage method and the number of clusters selected.

Sensitivity analysis

A sensitivity analysis was conducted on HDBSCAN, the best-performing clustering model, to examine how its parameter settings affected the association between complaint clusters and subsequent recalls. The analysis was first performed on a 150,000-record sample, using a baseline recall rate of 43.9%.

[Table 9: Unsupervised Learning HDBSCAN Model Sensitivity Results]

min_cluster_size	min_samples	clusters	noise	top recall rate (n)	clusters ≥70%	lowest recall rate
200	10	122	38.3%	92.2% (230)	12 (9.8%)	4.6%
200	25	117	34.9%	91.8% (231)	10 (8.5%)	4.4%
500	10	62	32.6%	89.4% (963)	4 (6.5%)	4.6%
500	25	59	31.9%	79.8% (1,771)	3 (5.1%)	4.4%
1000	10	31	26.9%	79.4% (1,803)	2 (6.5%)	18.8%
1000	25	34	28.7%	79.8% (1,771)	2 (5.9%)	18.3%

Across all parameter settings, HDBSCAN identified clusters with recall rates between 79% and 92%, approximately 1.8 to 2.1 times the baseline. However, the strength of the highest recall rate decreased as the minimum cluster size increased. When the minimum cluster size was set to 200, the highest recall rate reached 92.2% for a cluster containing 230 complaints. At a minimum cluster size of 1,000, the highest rate decreased to approximately 79% for clusters containing around 1,800 complaints.

The analysis also showed that the min_samples parameter could affect recall findings even when its impact on internal clustering results was limited. For a minimum cluster size of 500, increasing min_samples from 10 to 25 changed the number of clusters only slightly, from 62 to 59, and the proportion of noise points from 32.6% to 31.9%. However, the highest recall rate decreased from 89.4% to 79.8%, while the corresponding cluster size increased from 963 to 1,771 complaints. This effect was not observed to the same extent at minimum cluster sizes of 200 or 1,000, suggesting that the influence of min_samples depends on the selected minimum cluster size.

Finally, the analysis demonstrated that dataset size substantially affected the results. Using the same parameter settings (min_cluster_size=500 and min_samples=25), the 150,000-record sample produced three clusters with recall rates of at least 70%, representing 5.1% of its clusters, and a maximum recall rate of 79.8%. In comparison, the full dataset of approximately 2.25 million complaints produced 45 clusters with recall rates of at least 70%, representing approximately 16% of its clusters, and a maximum recall rate of 96.2%. Therefore, although the sensitivity analysis provides insight into how parameter choices influence clustering outcomes, the final reported results are based on the full dataset rather than the smaller tuning sample.

Discussion

Part A: Supervised Learning

Various findings surprised us in supervised learning. First, K-nearest neighbors (KNN) outperformed the other evaluated models on most ranking metrics. This was notable because KNN is a relatively simple, distance-based method, while random forest can capture more complex nonlinear relationships through an ensemble of decision trees. Another unexpected result was that hyperparameter tuning did not improve KNN's performance, suggesting that the default configuration was already reasonably effective. Random forest, in contrast, showed a larger difference between validation and test performance, with PR-AUC decreasing from 0.241 to 0.181 after tuning. These findings highlighted that model complexity and hyperparameter tuning do not necessarily lead to better predictive performance, particularly when the available features provide only a limited signal for identifying future recalls.

One of the main challenges was the imbalance in the target variable. Only 10.2% of complaints in the test set were followed by a matching recall within two years, making accuracy an unsuitable primary metric. Further, while KNN achieved the strongest overall ranking performance, its relatively low precision meant that it generated many false positives, limiting its real-world applicability. Finally, identifying the reasons behind false positives and false negatives was difficult because consumer complaints do not capture every source of information used to identify defects. Some recalls may result from manufacturer investigations or other sources, while complaints involving newly introduced vehicles may lack sufficient historical examples for the model to identify similar cases.

Several extensions could improve the predictive performance and practical usefulness of the supervised learning pipeline. First, the sensitivity analysis could be expanded to evaluate how performance changes under different prediction horizons, component classification levels, and training sample sizes. This would help determine whether the model's performance is consistent across different definitions of a future recall and whether additional training data produces meaningful improvements. Further, incorporating additional data sources, such as vehicle production volumes, warranty claims, manufacturer investigations, and service records, could provide information that is not available in consumer complaints alone. These improvements could help distinguish complaints that reflect isolated problems from those that indicate broader defects and ultimately make the model more useful.

Part B: Unsupervised Learning

Various findings surprised us in unsupervised learning. First, adding structured features, including severity flags, vehicle make, and model year, reduced clustering quality compared with using sentence embeddings alone. This differed from the original proposal, which anticipated that combining complaint text with vehicle metadata could improve the clustering results.

There were challenges as well. The quadratic memory complexity of agglomerative clustering was one of the biggest computational challenges, as it made training on the full dataset infeasible. We therefore limited this model to a sample of 50,000 complaints and checked whether the sample size substantially affected performance. HDBSCAN also presented computational challenges during full-dataset training, with an initial run taking more than an hour.

Several extensions could improve the analysis and provide a more comprehensive comparison of the clustering methods. First, feature selection could be performed separately for each model. In the current workflow, Stage 0 used K-means to select the final feature representation, which was then applied to HDBSCAN and agglomerative clustering. However, density-based and hierarchical methods may respond differently to structured features such as severity flags, vehicle make, and model year. Further, HDBSCAN tuning could be repeated on the full dataset, with a broader range of min_samples values beyond 25, to determine whether the selected parameters remain appropriate at scale.

Ethical Considerations

While this project uses publicly available data, it would be remiss to assume that there are no ethical concerns. In particular, the project raises considerations related to public safety, privacy, and transparency.

First, for Part A, which uses supervised learning to identify patterns in vehicle complaints and recalls, model predictions should not be treated as an automatic call to action. Inaccurate predictions could lead to unnecessary investigations that divert resources from more pressing safety concerns, cause reputational harm to manufacturers, or create false reassurance about potentially unsafe vehicles. Therefore, model outputs should be interpreted as preliminary indicators rather than conclusive findings.

Second, for Part B, which uses unsupervised learning to identify natural groupings in vehicle complaints, privacy is an important consideration. Although the data is publicly available, complaint narratives may contain personally identifiable information, including VIN and other details that could be linked to individuals. Any project using these datasets should minimize the collection, use, and disclosure of identifying information wherever possible.

Overall, these considerations highlight the importance of transparency regarding data limitations, cautious interpretation of model predictions, and responsible communication of results.

Statement of Work

Although individual team members took the lead on specific sections of the project, everyone contributed to nearly every aspect of the work.

Task	ME	CHF	JP

Project Proposal	Responsible	Support	Support
Data Collecting	Support	Responsible	Support
Revised Project Proposal	Support	Support	Responsible
Data Cleaning / Merging	Support	Responsible	Support
Data Manipulation / Feature Engineering	Responsible	Support	Support
Exploratory Data Analysis (EDA)	Support	Support	Responsible
Part A: Supervised Modeling	Responsible	Support	Support
Part B: Unsupervised Modeling	Support	Responsible	Support
Data Visualization	Responsible	Responsible	Responsible
Data Validation / Quality Checks	Support	Support	Responsible
Project Report	Responsible	Responsible	Responsible

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